University of Ottawa

Department of Mathematics and Statistics

MAT 1341A: Introduction to Linear Algebra

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Final Exam – Solutions

December 2025

Surname _____ First Name _____

Student # _____ Seat # _____

Instructions:

(a) You have 3 hours to complete this exam.

(b) This exam consists of two parts. Questions 1 through 13 are *multiple choice*. For these questions, only your answer in the designated table will be considered for marks. Questions 14 through 18 are *long answer*. For these questions, you must show your work and justify your answers to receive full marks. Partial marks may be awarded for making sufficient progress towards a solution.

(c) The number of points available for each question is indicated in square brackets.

(d) All work to be considered for grading should be written in the space provided. The reverse side of pages is for scrap work. If you find that you need extra space in order to answer a particular question, you should continue on the reverse side of the page and indicate this *clearly*. Otherwise, the work written on the reverse side of pages will not be considered for marks.

(e) Write your student number at the top of each page in the space provided.

(f) You may use the the blank pages at the end of the exam as extra space for solutions. If you do so, indicate clearly on the page of the relevant question that you have continued your solution on the extra page.

(g) You may also use the extra blank pages at the end of the exam as scrap paper.

(h) Cellular phones, unauthorized electronic devices or course notes are not allowed during this exam. Phones and devices must be turned off and put away in your bag. Do not keep them in your possession, such as in your pockets. If caught with such a device or document, the following may occur: you will be asked to leave immediately the exam and academic fraud allegations will be filed which may result in you obtaining a 0 (zero) for the exam.

By signing below, you acknowledge that you are complying with statement (h) above:

Signature: _____

Good luck!

Please do not write in the table below.

Question	1–13	14	15	16	17	18	Total
Maximum	13	3	7	3	3	3	32
Grade							

Write down your answers for Questions 1-13 in the table below.

Question	Your Answer
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	
13	

Part A: Answer Only Questions

For each of the Questions 1–13, **Write your answer (exactly one of the letters A,B,C,. . .) in the designated space on the answer sheet (on page 2)**. Answers written elsewhere will not be graded.

1. **[1 point]** Let $A = \begin{bmatrix} 1 & 1 \\ 3 & 1 \end{bmatrix}$ and let B be a 2×3 matrix such that $A^T B = \begin{bmatrix} 1 & -1 & 3 \\ 3 & -2 & 7 \end{bmatrix}$. What is the second row of the matrix B?

A $\begin{bmatrix} 4 & -\frac{5}{2} & 9 \end{bmatrix}$

B $\begin{bmatrix} 4 & \frac{3}{2} & 9 \end{bmatrix}$

C $\begin{bmatrix} -1 & \frac{1}{2} & -2 \end{bmatrix}$

D $\begin{bmatrix} -\frac{5}{2} & -\frac{1}{2} \end{bmatrix}$

E $\begin{bmatrix} 4 & -1 \end{bmatrix}$

F $\begin{bmatrix} -4 & -\frac{5}{2} & 9 \end{bmatrix}$

Solution: C

2. **[1 point]** Suppose that A is a 4×7 matrix and B is a 6×4 matrix. Which of the statements I–V given below are true?

I. The columns of B can never form a basis of $\mathbb{R}^6$.

II. In every echelon form of A, the entries of the last row are all zeros.

III. If $\text{rank}(A) = 4$, then for every $\mathbf{b} \in \mathbb{R}^4$ the linear system $A\mathbf{x} = \mathbf{b}$ is consistent.

IV. If $\text{rank}(B) = 4$, then for every $\mathbf{b} \in \mathbb{R}^6$ the linear system $B\mathbf{x} = \mathbf{b}$ is consistent.

V. $\text{Nul}\,(BA)$ is a subspace of $\mathbb{R}^6$.

Select the correct answer:

A I and IV.

B II and V.

C II and III.

D I and III

Solution: D

3. **[1 point]** Let V be the vector space of polynomials of degree at most 4 in one variable x, with coefficients in $\mathbb{R}$. What is the dimension of the following subspace W of V?

$$W = \text{Span}\left\{x - x^3, 2x + 3x^2 - 2x^3, -3x + x^2 + 3x^3, x + 2x^2 - x^3\right\}$$

A $\dim W = 1$.

B $\dim W = 2$.

C $\dim W = 3$.

D $\dim W = 4$.

Solution: *B*

4. **[1 point]** Which of the following sets is a subspace of the given vector space V?

I. $\left\{ A \in \text{Mat}_{3\times 2}(\mathbb{R}) \;\middle|\; A^T A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right\}$, $V = \text{Mat}_{3\times 2}(\mathbb{R})$.

II. $\left\{ \begin{bmatrix} x \\ x+y \\ z-2x \\ t+y \end{bmatrix} \;\middle|\; x, y, z, t \in \mathbb{R} \right\}$, $V = \mathbb{R}^4$.

III. $\left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \;\middle|\; x, y, z \in \mathbb{R} \text{ and } x + y \geq 0 \right\}$, $V = \mathbb{R}^3$.

IV. $\left\{ f \in V \;\middle|\; f(10) = 10 \right\}$, V = polynomials in one variable x with coefficients in $\mathbb{R}$.

V. $\left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \;\middle|\; x_1, x_2, x_3 \in \mathbb{R} \text{ and } \begin{bmatrix} 3 & 1 & 3 \\ 1 & 1 & -1 \\ 1 & 2 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2x_1 \\ 2x_2 \\ 2x_3 \end{bmatrix} \right\}$, $V = \mathbb{R}^3$.

Select the correct answer:

A I and V.

B II and V.

C II and IV.

D I and III

E III and IV.

Solution: B

5. **[1 point]** Let A be an $n \times n$ matrix. Which of the assertions I–VI given below is NOT equivalent to the statement "A is invertible"?

I. The rows of A are linearly independent.

II. $\text{Nul}(A)^{\perp} = \mathbb{R}^n$.

III. $\text{Col}(A^T) = \{0\}$.

IV. $\det(A) \neq 0$.

V. The reduced echelon form of A is an $n \times n$ diagonal matrix.

VI. The linear system $A\mathbf{x} = \mathbf{b}$ is consistent for every vector $\mathbf{b} \in \mathbb{R}^n$.

Solution: D

Select the correct answer:

A I and V.

B II and VI.

C II and V.

D III and V.

6. **[1 point]** Suppose that A, B, and C are 4×4 matrices such that $\det(A) = -2$, $\det(B) = 5$, and $\det(-A^T BCB^{-2}) = 3$. Which of the following is $\det(C)$?

A $\frac{15}{2}$.

B $-\frac{15}{2}$.

C $\frac{5}{6}$.

D $-\frac{5}{6}$.

Solution: *B.*

7. **[1 point]** The vectors $\mathbf{v}_1 = (1, 0, 1, 2)$, $\mathbf{v}_2 = (5, 0, -1, -2)$, and $\mathbf{v}_3 = (0, 1, -2, 1)$ are orthogonal. If we write $\mathbf{v} = (-9, 3, -3, 9)$ as a linear combination $\mathbf{v} = a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + a_3\mathbf{v}_3$, then what is the value of a_2?

A $a_2 = 3$.

B $a_2 = -3$.

C $a_2 = 2$.

D $a_2 = -2$.

Solution: *D.*

8. **[1 point]** Consider the matrix

$$A = \begin{bmatrix} 1 & -3 & 4 & 6 \\ -\frac{1}{2} & t & 1 & 2 \\ 1 & -3 & 1 & 1 \\ 2 & -6 & 8 & 12 \end{bmatrix}.$$

Answer the following three (3) questions in this order:

- For which value of t do we have $\text{rank}(A) = 2$?
- If $t = 2$, what is $\dim \text{Nul}(A)$?
- If $t = 2$, are the columns of A linearly dependent (LD) or linearly independent (LI)?

Select the correct answer:

A $t = -\frac{3}{2}$, $\dim \text{Nul}(A) = 2$, LI.

B $t = \frac{1}{2}$, $\dim \text{Nul}(A) = 2$, LD.

C $t = \frac{3}{2}$, $\dim \text{Nul}(A) = 1$, LD.

D $t = -\frac{1}{2}$, $\dim \text{Nul}(A) = 1$, LI.

E $t = \frac{3}{2}$, $\dim \text{Nul}(A) = 2$, LI.

F $t = \frac{1}{2}$, $\dim \text{Nul}(A) = 1$, LI.

Solution: C

9. **[1 point]** Let $z = 1 - 2i$ and $w = i + 2$. Simplify the complex number $\dfrac{w}{1 + \overline{z}}$ in the form $a + bi$ where a and b are real numbers.

A $\frac{3}{4} + \frac{1}{4}i$.

B $\frac{3}{2} - \frac{1}{2}i$.

C $\frac{3}{4} - \frac{1}{4}i$.

D $\frac{3}{2} + \frac{1}{2}i$.

E $-\frac{3}{4} + \frac{1}{4}i$.

Solution: C

10. **[1 point]** Let A be a 5×5 matrix with entries in $\mathbb{R}$. Suppose that the characteristic polynomial of A is

$$-(\lambda - 2)(\lambda^2 + 2\lambda + 3)(\lambda + 1)^2.$$

Which of the assertions I–IV below are <u>true</u>?

I. A is invertible.

II. There exists a nonzero vector $\mathbf{v} \in \mathbb{R}^5$ such that $A\mathbf{v} = 3\mathbf{v}$.

III. $\dim(\text{Nul}(A + I_5)) \leq 2$. Here I_5 denotes the 5×5 identity matrix.

IV. The equation $A^T\mathbf{x} = \mathbf{x}$ has a nonzero solution $\mathbf{x} \in \mathbb{R}^5$.

Select the correct answer:

A I and III.

B I and IV.

C II and III.

D III and IV.

Solution: A

11. **[1 point]** Let $T : \mathbb{R}^3 \to \mathbb{R}^2$ be the linear transformation defined by

$$T(x, y, z) = (x + 2z, -3x + y + z).$$

Which of the statements I–IV below are <u>true</u>?

I. The standard matrix of T has 3 rows and 2 columns.

II. the vector $(2, 7, -1)$ is in $\text{Ker}(T)$.

III. $\ker(T)$ is a subspace of $\mathbb{R}^3$.

IV. $\dim(\text{Im}(T)) = 3$.

Select the correct answer:

A I and II.

B I and III.

C III and IV.

D II and III.

Solution: D

12. **[1 point]** Let $A = \begin{bmatrix} 1+i & i \\ -1 & 0 \end{bmatrix}$. Find the value of z such that the vector $\begin{bmatrix} 1-i \\ z \end{bmatrix}$ is an eigenvector of A with eigenvalue i.

A $z = -1 + i$.

B $z = -i$.

C $z = 1 + i$.

D $z = -1 - i$.

Solution: *C*

13. **[1 point]** Let V be the vector space of polynomials in one variable x, with coefficients in $\mathbb{R}$. Let

$$U = \{f \in V \mid f(0) = f(1) - 1\}.$$

Which of the following statements are true?

I. U does not contain the zero vector.

II. U is closed under scalar multiplication, that is, if $f \in U$ and $r \in \mathbb{R}$, then $rf \in U$.

III. U is closed under addition, that is, if $f, g \in U$, then $f + g \in U$.

IV. $x^3 - x^2 + x + 1 \in U$.

Select the correct answer:

A I and IV.

B II and IV.

C II and III.

D III and IV.

Solution: A

Part B: Long Answer Questions

For Questions 14–18, you must show your work and justify your answers to receive full marks. Partial marks may be awarded for making sufficient progress towards a solution.

14. **[3 points]** Determine the value of the parameter t such that the following linear system is inconsistent.

$$\begin{cases} -x_1 + 3x_2 - 3x_3 + x_4 = 2 \\ \quad 2x_2 + x_3 - x_4 = 5 \\ x_1 - 3x_2 - tx_3 - x_4 = -1 \end{cases}$$

Solution: The augmented matrix of the system can be reduced to an echelon form as follows:

$$\left[\begin{array}{cccc|c} -1 & 3 & -3 & 1 & 2 \\ 0 & 2 & 1 & -1 & 5 \\ 1 & -3 & -t & -1 & -1 \end{array}\right] \xrightarrow{R_3 \to R_3 + R_1} \left[\begin{array}{cccc|c} -1 & 3 & -3 & 1 & 2 \\ 0 & 2 & 1 & -1 & 5 \\ 0 & 0 & -t-3 & 0 & 1 \end{array}\right]$$

The system is inconsistent if and only if $-t - 3 = 0$, i.e., $t = -3$.

15. (a) **[4 points]** Let $A = \begin{bmatrix} 4 & 2 & -1 \\ 1 & 4 & 0 \\ -1 & 0 & 2 \end{bmatrix}$. Compute a basis for the eigenspace of A corresponding to the eigenvalue $\lambda = 3$.

Solution: We have $A - \lambda I = A - 3I = \begin{bmatrix} 1 & 2 & -1 \\ 1 & 1 & 0 \\ -1 & 0 & -1 \end{bmatrix}$. We reduce to REF:

$$\begin{bmatrix} 1 & 2 & -1 \\ 1 & 1 & 0 \\ -1 & 0 & -1 \end{bmatrix} \xrightarrow{R_2 \to R_2 - R_1} \begin{bmatrix} 1 & 2 & -1 \\ 0 & -1 & 1 \\ -1 & 0 & -1 \end{bmatrix} \xrightarrow{R_3 \to R_3 + R_1} \begin{bmatrix} 1 & 2 & -1 \\ 0 & -1 & 1 \\ 0 & 2 & -2 \end{bmatrix}$$

$$\xrightarrow{R_3 \to R_3 + 2R_2} \begin{bmatrix} 1 & 2 & -1 \\ 0 & -1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_2 \to -R_2} \begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 \to R_1 - 2R_2} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

The corresponding linear system is $\begin{cases} x + z = 0 \\ y - z = 0 \end{cases}$ Now x and y are basic and z is free. Thus the general solution is

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -z \\ z \\ z \end{bmatrix} = z \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$$

Hence the basis for the eigenspace is $\left\{ \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} \right\}$.

(b) **[2 points]** Let $B = \begin{bmatrix} -7 & 0 & 4 \\ 2 & -1 & -2 \\ -8 & 0 & 5 \end{bmatrix}$. Compute the characteristic polynomial of B.

Solution: We use cofactor expansion with respect to the second column. We obtain

$$\det(B-\lambda I) = \begin{vmatrix} -7-\lambda & 0 & 4 \\ 2 & -1-\lambda & -2 \\ -8 & 0 & 5-\lambda \end{vmatrix} = (-1-\lambda)\big((-7-\lambda)(5-\lambda)+32\big) = (-1-\lambda)(\lambda^2+2\lambda-3) = -(\lambda+1)(\lambda-1)(\lambda+3).$$

(c) **[1 point]** Is the matrix B of part (b) diagonalizable? You should justify your answer.

Solution: The matrix is diagonalizable because it has 3 distinct eigenvalues.

16. **[3 points]** Using the Gram-Schmidt method, find an orthogonal basis for the subspace W of $\mathbb{R}^4$ that is spanned by

$$\mathbf{v}_1 = (1,1,0,-1)\ ,\ \mathbf{v}_2 = (-2,1,1,1)\ ,\ \mathbf{v}_3 = (-1,2,1,0)$$

Solution: First we show that these vectors are linearly independent. Forming a matrix with $\mathbf{v}_1, \mathbf{v}_2$, we obtain

$$\begin{bmatrix} 1 & -2 & -1 \\ 1 & 1 & 2 \\ 0 & 1 & 1 \\ -1 & 1 & 0 \end{bmatrix} \xrightarrow{R_2 \to R_2 - R_1} \begin{bmatrix} 1 & -2 & -1 \\ 0 & 3 & 3 \\ 0 & 1 & 1 \\ -1 & 1 & 0 \end{bmatrix} \xrightarrow{R_4 \to R_4 + R_1} \begin{bmatrix} 1 & -2 & -1 \\ 0 & 3 & 3 \\ 0 & 1 & 1 \\ 0 & -1 & -1 \end{bmatrix}$$

$$\xrightarrow{R_3 \to \frac{1}{3}R_2} \begin{bmatrix} 1 & -2 & -1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & -1 & -1 \end{bmatrix} \xrightarrow[R_3 \to R_3 - R_2]{R_4 \to R_4 + R_2} \begin{bmatrix} 1 & -2 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

The first two columns are pivot columns. Thus, $\mathbf{v}_1, \mathbf{v}_2$ forms a basis for $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$.

Now we apply the Gram-Schmidt process to $\{\mathbf{v}_1, \mathbf{v}_2\}$. We set

$$\mathbf{b}_1 = \mathbf{v}_1 = (1,1,0,-1)$$

and

$$\mathbf{b}_2 = \mathbf{v}_2 - \text{Proj}_{\mathbf{v}_1}\mathbf{v}_2 = \mathbf{v}_2 - \frac{\mathbf{v}_2 \bullet \mathbf{v}_1}{\mathbf{v}_1 \bullet \mathbf{v}_1}\mathbf{v}_1 = (-2,1,1,1) - \frac{-2}{3}(1,1,0,-1) = (-\frac{4}{3}, \frac{5}{3}, 1, \frac{1}{3}).$$

17. **[3 points]** Find the best approximation to a solution for the linear system

$$\begin{cases} x + y = 3 \\ x + 3y = 5 \\ 2x - 2y = -2 \end{cases}$$

Solution: The system is of the form $A\mathbf{x} = \mathbf{b}$ where

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 3 \\ 2 & -2 \end{bmatrix} \quad , \quad \mathbf{b} = \begin{bmatrix} 3 \\ 5 \\ -2 \end{bmatrix}$$

For best approximation, note that

$$A^T A = \begin{bmatrix} 6 & 0 \\ 0 & 14 \end{bmatrix} \quad , \quad A^T \mathbf{b} = \begin{bmatrix} 4 \\ 22 \end{bmatrix}$$

Thus we have the linear system $A^T A\mathbf{x} = A^T \mathbf{b}$ which is

$$\begin{bmatrix} 6 & 0 \\ 0 & 14 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 22 \end{bmatrix} \Rightarrow \begin{cases} 6x = 4 \\ 14y = 22 \end{cases}$$

Thus

$$x = \frac{2}{3} \quad , \quad y = \frac{11}{7}$$

18. **[3 points]** Compute the following determinant

$$\begin{vmatrix} -\frac{1}{2} & 1 & 3 & 5 \\ -1 & 3 & 7 & 8 \\ 0 & 0 & 5 & -3 \\ \frac{1}{2} & -1 & -3 & -1 \end{vmatrix}$$

Using the row reduction method. You should not use the cofactor expansion method.

Solution:

$$\begin{bmatrix} -\frac{1}{2} & 1 & 3 & 5 \\ -1 & 3 & 7 & 8 \\ 0 & 0 & 5 & -3 \\ \frac{1}{2} & -1 & -3 & -1 \end{bmatrix} \xrightarrow[R_4 \to R_4 + R_1]{R_2 \to R_2 - 2R_1} \begin{bmatrix} -\frac{1}{2} & 1 & 3 & 5 \\ 0 & 1 & 1 & -2 \\ 0 & 0 & 5 & -3 \\ 0 & 0 & 0 & 4 \end{bmatrix}$$

The last matrix is upper triangular, thus its determinant is

$$(-\frac{1}{2})(1)(5)(4) = -10.$$

Since the process did not use any interchanges, we obtain

$$\begin{vmatrix} -\frac{1}{2} & 1 & 3 & 5 \\ -1 & 3 & 7 & 8 \\ 0 & 0 & 5 & -3 \\ \frac{1}{2} & -1 & -3 & -1 \end{vmatrix} = -10.$$